University of North Carolina at Charlotte
Department of Electrical and Computer Engineering
ECGR 3120 Inkjet Printer Simulation

Group Project Report

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Q1

Simulate the motion of the droplet when no voltage is applied across the capacitor. How much time would it take for the droplet to reach the center of the paper?

The droplet is travelling along the y-axis towards the paper. Because there is no acceleration in the z-direction:

$$V_Z = 20m/s = 20\ mm/ms\ (constant) \text{ which is independent of } V_Y.$$

The path taken by each droplet regardless of the voltage applied is the same in the x direction. This displacement can be calculated by integrating V_Z which gives us:

$$S_Z = 20 * t\ mm$$

Solving for time gives us: $\frac{S_Z}{20} = t\ sec$. It is given that the distance from droplet gun to the paper is 3mm.

Therefore, the time taken by a droplet to reach the center of the paper with no voltage applied across the capacitor is **0.15 milliseconds**.

Q2

Simulate the act of drawing a vertical line (similar to the letter "I") along the center of the paper at 300 dpi resolution. How much time would it take to draw the letter "I"?

The time it takes for the letter **I**, to be drawn depends on two factors:

- Length of the letter
- Printer's firing rate(T_F)

This is assuming the distance to the paper(D), and V_Z are constant. The length of the letter determines the number of droplets (N) required to draw it. To keep the units we are using uniform we are converting the given resolution from inches to meters:

$$\frac{300\ dots}{1in} * \frac{1in}{25.4mm} = \boxed{11.81 \text{ dpmm}}$$

This means that for a 1mm long line, the printer would require 11.81 dots. For the simulation, this isn't possible since the number of dots has to be an integer value, which means we round up to **12 dots**. From here,

$$\text{Time for the first dot to hit the paper} = \frac{D}{V_X}$$

The remaining **N - 1** droplets are spaced equally from each other due to the constant firing rate and will hit the paper by the time the next droplet is fired, so the remaining time is: $(N-1) * T_F$

$$\text{Therefore, } T_{total} = \frac{D}{V_X} + (N-1) * T_F$$

Moving on, we need to print "I" in the shortest possible time, to do so we need to decide on the fastest possible firing rate. T_F is dependent on the capacitors as they cannot have more than one droplet passing through them at a time as each droplet needs to be influenced by a different amount of voltage applied to the capacitors. This implies that:

$$T_{F,max} = T_C = \frac{L_1}{V_X} = 25\ \mu s = 0.025\ ms$$

Here, T_C is the time spent by each droplet inside the capacitor.
We know have all the parts required to find the total time required to the draw a 1mm long vertical line across the paper:

$$T_{total} = \frac{D}{V_X} + (N-1) * T_C = \frac{3\ mm}{20\ mm/ms} + (12-1) * 0.025\ ms$$

$$\boxed{\therefore T_{total} = 0.425\ ms}$$

Q3

Plot the profile of applied voltage V(t) across the capacitor versus the time t for drawing the letter 'I'. This profile of V(t) should look like a staircase. V(t) will remain constant until a droplet exits the capacitor chamber, and immediately it will switch to the next value in order to control the next incoming droplet. How big the letter 'I' could be drawn on the paper?

The largest letter "I" that can be drawn is 6mm, this is due to the maximum deflection within the capacitor being 0.5m. If you set the y displacement formula equal to 0.5:

$$Y_{C,max} = 0.5mm = \frac{q*E*t^2}{2m}; E = \frac{V}{w}; \text{ w = width between capacitors}$$
$$\boxed{V_{max} = \frac{(0.5\ mm)*2*m*w}{q*t^2} = 2610.11\ V}$$

This voltage value gives us the exit velocity of the droplet:

$$V_{exit} = \frac{q*E_{max}*T_C}{2*m} = 40\ m/s$$
$$\therefore Y_{disp,max} = 2.5\ mm$$
$$\boxed{\therefore \text{Biggest "I"} = 6\text{mm}}$$

Q4

X

Resimulate the act of drawing letter 'I' as big as possible consuming as small time as possible for each of the following adjustment of the parameter:

a) The distance between the capacitor and the paper L2 is threefold increased, and everything else remains same

$$L_{2,new} = 3 * L_2 = 3.75\ mm$$
$$\text{Time from capacitor to paper} = T_{3,new} = \frac{3.75\ mm}{20\ mm/ms} = 0.1875\ ms$$
$$Y_{disp,max,new} = V_{exit} * T_{3,new} = 40 * 0.1875 = 7.5\ mm$$
$$\boxed{\therefore Biggest\ "I" = 16\ mm}$$

b) L1 is twofold increased, and everything else remains same

$$L_{1,new} = 2 * L_1 = 1\ mm$$
$$\Rightarrow T_{F,max} = T_C = \frac{1\ mm}{V_X} = 50\ \mu s$$
$$wkt,\ Y_{C,max} = 0.5\ mm = \frac{q*E*T_{C,new}}{2*m}$$
$$\Rightarrow V_{Max,new} = \frac{(0.5\ mm)*2*m*w}{q*t^2} = 652.6\ V$$
$$\Rightarrow V_{exit} = \frac{q*E_{max}*T_{C,new}}{m} = 20\ m/s$$
$$\Rightarrow Y_{d,max} = 1.25\ mm$$
$$\boxed{\therefore Biggest\ "I" = 3.5\ mm}$$

c) Droplet diameter is tenfold increased, and everything else remains same

$$d_{new} = 2 * d = 168\ \mu m$$
$$\Rightarrow m_{new} = \rho * \frac{4*\pi*(d/2)^3}{3} = 2.483 * 10^{-9}\ kg$$
$$\Rightarrow V_{Max,new} = \frac{(0.5\ mm)*2*m_{new}*w}{q*t^2} = 20,907\ V$$
$$\Rightarrow V_{exit} = \frac{q*E_{max}*T_C}{m_{new}} \approx 40\ m/s$$
$$\Rightarrow Y_{disp,max} = 2.5\ mm$$
$$\boxed{\therefore Biggest\ "I" = 6\ mm}$$

d) The horizontal speed at which the gun shoots the droplet is twofold increased, and everything else remains same

$$V_{X,new} = 2 * V_X = 40\ mm/ms$$
$$\therefore T_{C,new} = \frac{L_1}{V_{X,new}} = \frac{0.5\ mm}{40\ mm/ms} = 12.5\ \mu s$$
$$\Rightarrow V_{Max,new} = \frac{(0.5\ mm)*2*m*w}{q*t^2} = 10,442.1\ V$$
$$\Rightarrow V_{exit} = \frac{q*E_{max}*T_{C,new}}{m} \approx 80\ m/s$$
$$\Rightarrow Y_{disp,max} = 2.5\ mm$$
$$\boxed{\therefore Biggest\ "I" = 6\ mm}$$

e) The charge of the droplet is fivefold increased, and everything else remains same.

$$q_{new} = 5 * q = -9.5 * 10^{-10}$$
$$\Rightarrow V_{Max,new} = \frac{(0.5\ mm)*2*m*w}{q_{new}*t^2} = -522.1\ V$$
$$\Rightarrow V_{exit} = \frac{q_{new}*E_{max}*T_C}{m} \approx 40\ m/s$$
$$\Rightarrow Y_{disp,max} = 2.5\ mm$$
$$\boxed{\therefore Biggest\ "I" = 6\ mm}$$

Q5

For our design paramters, we used the same values as before because any changes to these values brought scaling issues in simulation due to the assymmetry between both pairs of capacitors. In addition, there are an excess of additional conditions needed to be programmed, further enforcing our decision to use the same parameters for these additional capacitors. Simulating this new setup gave us an "H" that had dimensions **6 mm x 6 mm**.

The links below will take you to our:

- Our Calculations
- GitHub Repo
- Video Demonstration